Prototype Reflection

How the Unity prototype connects to the Kast concept document

The prototype

The prototype is a single Unity level built with the Universal Render Pipeline. The player must get a physics cube, and then themselves, onto a goal platform that is too high to reach directly. Two spells are available: **Levitate**, which makes the targeted cube rise continuously for a few seconds, and **Swap**, which exchanges the player's position with the cube's. The puzzle is solved by levitating the cube up to the high platform, then swapping with it to arrive there.

How it reflects the concept

- **Laboratory setting.** The level is dressed as a test room in the Institute, and whiteboards introducing Levitate and Swap stand in for the chalkboard-covered lab notes described in the concept document, giving the prototype the same in-world, diegetic style of teaching a new glyph.
- **Draw it, don't select it.** Levitate and Swap are each triggered by drawing a distinct glyph with the mouse rather than by a keybind or menu, which is the concept document's first design pillar and its main point of difference from other physics puzzlers.
- **Core loop.** The prototype follows exactly the micro and puzzle loops described in the concept document: the player targets an object, casts a spell, watches the physics react, and reasons about the result before acting again. Reaching the goal door on an unreachable platform is a direct, minimal instance of the "use spells on physics objects to reach the exit" premise the whole game is built on.
- **Physics you can trust.** Levitate's rise is a predictable, repeatable behaviour, and Swap is a clean, deterministic position exchange. Both stick to the pillar that every solution should come from understanding a consistent rule, not from luck or reflexes.
- **One new idea, then combined.** The two-spell structure mirrors the concept's Teach–Test–Twist–Combine pattern at the smallest possible scale: Levitate and Swap can each be understood alone, but the level only opens once the player realises they must be used *together*, in a specific order. This is the same design move used throughout the full spell roster in the concept document (for example Lift into Halt to build a hover platform).
- **First-person, physics-object puzzle solving.** The prototype keeps the concept's first-person camera and its focus on manipulating a physics object rather than the environment itself, which is the throughline that connects Kast to its main inspiration, Portal.
- **Fair and forgiving.** Because Levitate and Swap are non-destructive and the cube cannot be lost from the level, the room can be reset and retried instantly, in line with the no-death, low-frustration approach described under Player Experience and POV.

How the prototype's spells fit the roster

Levitate and Swap are two of the spells in the concept document's roster, so this first level is a direct, working sample of that roster rather than a stand-alone pair. Levitate covers the vertical-movement role, and Swap contributes a distinct verb, exchanging positions, that broadens what the roster as a whole can do. Building the prototype around these two first was a useful way to test the "combine two verbs to reach new space" idea before committing art and level time to the rest of the set.

The prototype implements mouse-drawn glyph casting: Levitate and Swap each have their own symbol, and the player casts by drawing the matching glyph, exactly as described under Casting System in the concept document. The level is set in a prototype laboratory environment, with whiteboards that introduce each of the two spells, in line with the Institute for Applied Glyphics setting and its chalkboard

lab notes described under Game World Fiction. What is not yet in this first level is the ink resource and Ink's own dialogue, which are planned for later passes.

Next steps

- Build the teaching rooms for the rest of the roster in the same whiteboard style used here for Levitate and Swap.
- Extend the whiteboard-note style of teaching a glyph to future levels, alongside the full tile, brass and chalkboard art direction described in Section 7.